

Mathematical Model of the MSW Recovery-Pathway Screening

All equations used in the study, as implemented in the open-source package mswpath v3.2: data harmonisation, waste characterisation, life-cycle inventory and costs of six options, decision analysis, uncertainty, sensitivity, scenario discovery, value of information, break-even targets and the projection to 2045

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How to read this document

- Equations are numbered in the order of the computation. Each section names the module and function that implements it, so every equation can be traced to code (mswpath/core.py unless stated otherwise).
- Everything is expressed per tonne of mixed MSW, wet weight, as received at the facility gate in 2025 (functional unit). Masses m_j are tonnes of fraction j per tonne of MSW.
- In the Monte Carlo analysis every parameter is a vector of draws; all equations hold draw by draw.
- Numerical values are in the tables of Section 16, generated from data/*.csv, which also hold source and evidence status.

1 Notation and system

Indices: j {food, garden, paper, wood, textile, rubber, plastic, metal, glass, other} = the ten model fractions; k {SL, S1, ..., S5} = options; i = Monte Carlo draw; p = carbon value (USD/t CO₂e). Options: SL sanitary landfill with gas collection and flare; S1 waste-to-energy (grate incineration); S2 refuse-derived fuel (RDF) co-processed in the nearest cement kiln; S3 anaerobic digestion (AD) of food separated from mixed waste; S4 PHB bioplastic from landfill gas; S5 integrated RDF + AD. Baselines: open dump (OD) and SL. Every option manages the whole tonne; residues go to a residue landfill (sanitary unless stated). For each option the model returns G_k (kg CO₂e/t), C_k (USD/t), fossil cumulative energy demand CED_k (MJ/t) and land take LU_k (m²/t), with feasibility flags.

2 Data harmonisation to the 2025 baseline (MSWModel.harmonise, tonnage; build_database_v3.py)

2.1 Tonnage, projected once

$$Q_{2025} = Q_t (1 + g)^{2025 - t}, \quad M_{2025}^d = M_t^d (1 + g)^{2025 - t}, \quad M_{2025}^n = M_t^n (1 + g)^{2025 - t} \quad (1)$$

Q_t is the reported tonnage (t/day) of year t , g the RIPS growth rate (or the default g_0 of the register), d and n the domestic and non-domestic streams. The exponent is zero when the tonnage already refers to 2025; the model asserts $Q_{\text{used}} = Q_{2025}$ of the database (single-projection guard).

2.2 Composition

$$S_j^{(\ell)} = \frac{x_j^{(\ell)}}{\sum_{j'} x_{j'}^{(\ell)}}, \quad f^{(\ell)} = \frac{100}{\sum_{j'} x_{j'}^{(\ell)}}, \quad S_j = \frac{M_{2025}^d S_j^{(d)} + M_{2025}^n S_j^{(n)}}{M_{2025}^d + M_{2025}^n} \quad (2)$$

$x_j^{(\ell)}$ is the wet-mass percentage of stream ℓ after mapping the RIPS categories to the model fractions (sum of the categories mapped to j); blank categories are absent (zero share), never imputed in the main case. Harmonisation rules (sampled priors): if organics are lumped, a share γ of food is moved to garden; if wood $\geq 10\%$ with no leaf category, a share γ of wood is moved to garden:

$$L: S_{\text{garden}} \leftarrow S_{\text{garden}} + \gamma S_{\text{food}}, \quad S_{\text{food}} \leftarrow (1 - \gamma) S_{\text{food}}; \quad W: S_{\text{garden}} \leftarrow S_{\text{garden}} + \gamma S_{\text{wood}}, \quad S_{\text{wood}} \leftarrow (1 - \gamma) S_{\text{wood}} \quad (3)$$

Scenario *glass_imputed*, only where glass is not reported, with an imputed share u_g :

$$S_j \leftarrow S_j (1 - u_g) \quad (j \neq \text{glass}), \quad S_{\text{glass}} \leftarrow S_{\text{glass}} + u_g \quad (4)$$

3 Waste characterisation (MSWModel.model)

Moisture as received w_j , dry matter $(1 - w_j)$, dry lower heating value h_j (legacy values multiplied by the calibration factor k_h , plastic by $k_{h,p}$), latent heat λ :

$$M = \sum_j s_j w_j, \quad H = \max \left[0, \sum_j s_j (1 - w_j) k_{h,j} h_j - \lambda M \right] \quad (\text{MJ/kg}) \quad (5)$$

Fossil CO₂ released when a mass vector m is burned (C_j dry carbon, φ_j fossil share):

$$E_{fos}(m) = 10^3 \frac{44}{12} \sum_j m_j (1 - w_j) C_j \phi_j \quad (\text{kg CO}_2/\text{t}) \quad (6)$$

Moisture enters once: DOC and C are on a dry basis and multiply dry mass (DOC_{wet} = DOC_{dry}(1-w)). In the Monte Carlo all fraction moistures move together through one common draw $u_w; w_j = F_{tri}^{-1}(u_w; w_{j,lo}, w_j, w_{j,hi})$.

4 Landfill module (open dump and sanitary landfill)

For a mass vector m sent to a landfill plus inert mass m_{in} (ash, digestate):

$$CH_4(m) = 10^3 k_D MCF F \frac{16}{12} \sum_j m_j (1 - w_j) DOC_j DOCf_j \quad (\text{kg CH}_4/\text{t}) \quad (7)$$

$$CH_4^{capt} = \eta CH_4 [SL], \quad CH_4^{emit} = (CH_4 - CH_4^{capt}) (1 - OX [SL]) \quad (8)$$

$$G_{LF} = CH_4^{emit} GWP_{CH_4} + a_{LF}(t), \quad t = \sum_j m_j + m_{in} \quad (9)$$

$$C_{LF} = t c_u, \quad c_u = c_{SL} \left(\frac{\max(Qt, 5)}{Q_{ref, SL}} \right)^{-e_{SL}} (SL), \quad c_u = c_{OD} (OD) \quad (10)$$

MCF = MCF_{SL} (managed anaerobic) or MCF_{OD}; [SL] = 1 for the sanitary landfill and 0 for the open dump (no collection, no cover oxidation); k_D is a multiplier on DOC·DOCf; DOCf of rubber is the gap parameter DOCf_R. The landfill baseline values are G_{SL} = G_{LF}(s), G_{OD} = G_{LF}(s) with OD settings; L₀ = CH₄(s) with MCF = 1 is reported for transparency.

5 Techno-economic building blocks

$$CRF = \frac{r(1+r)^n}{(1+r)^n - 1}, \quad c^{cap}(K_{ref}, q_{ref}, q) = \frac{CRF K_{ref} (q/A/q_{ref})^b}{365 q} \quad (\text{USD/t}) \quad (11)$$

q = throughput (t/day) of the unit, A = availability (nameplate = throughput/A), b = capacity exponent, K_{ref} the CAPEX at reference capacity q_{ref}. Electricity is credited at the grid factor EF = EF_{grid} k_{EF} and valued at p_{el} (or the Perpres 109/2025 tariff in that scenario where Q ≥ 1,000 t/day).

6 The six options

6.1 SL sanitary landfill with flare

$$G_{SL} = G_{LF}(s), \quad C_{SL} = C_{LF}(s) \quad (12)$$

6.2 S1 waste-to-energy

$$E_{el} = \frac{10^3}{3.6} H \eta_W \quad (\text{kWh/t}), \quad G_1 = E_{fos}(s) + n_{N_2O} GWP_{N_2O} + a_W - E_{el} EF + G_{LF}(0, \alpha) \quad (13)$$

$$C_1 = c^{cap}(K_W, q_W, Q) + o_W - E_{el} p_{el} + C_{LF}(0, \alpha) \quad (14)$$

α = bottom ash and APC residue (t/t) landfilled as inert mass.

6.3 S2 RDF to cement kiln

$$r_j = s_j \min(\tau_j k_{\tau}, 1), \quad m_{in} = \sum_j r_j, \quad d = \sum_j r_j (1 - w_j), \quad m_{out} = \min(m_{in}, \frac{d}{1-\omega}) \quad (15)$$

$$e = \left[\sum_j r_j (1 - w_j) k_{h,j} h_j - \lambda (m_{out} - d) \right] k_{NCV}, \quad e_{net} = e - (m_{in} - m_{out}) q_{dry}, \quad m_{del} = m_{out} \frac{e_{net}}{e} \quad (16)$$

$$NCV_{RDF} = e/m_{out}, \quad G_2 = E_{fos}(r) + e_R EF + a_R + m_{del} Def_{tr} - \psi e_{net} EF_{coal} + G_{LF}(s - r) \quad (17)$$

$$C_2 = c^{cap}(K_R, q_R, Q) + o_R + m_{del} D c_{tr} - e_{net} p_{RDF} + C_{LF}(s - r) \quad (18)$$

τ_j transfer coefficients to RDF, ω target moisture, q_{dry} dryer heat per t of water, ψ GJ coal displaced per GJ RDF, D = road distance to the nearest kiln = great-circle distance $\times$ tortuosity, $k_{NCV} = 1$ (0.78 in the stress test).

6.4 S3 anaerobic digestion of food separated from mixed waste

$$a = \kappa S_{food}, \quad V = 10^3 a (1 - w_{food}) \frac{VS}{TS} y_{CH_4} k_{mech} \text{ (Nm}^3\text{/t)}, \quad E_{AD} = V(1 - f_{AD}) \frac{LHV_{CH_4}}{3.6} \eta_{CHP} (1 - \pi_{AD}) \quad (19)$$

$$G_3 = V \rho_{CH_4} f_{AD} GWP_{CH_4} + a a_A - E_{AD} EF + G_{LF}(s - a e_{food}, \delta a) + \pi(e_R EF + a_R) \quad (20)$$

$$C_3 = a[c^{cap}(K_A, q_A, Qa) + o_A + c_{pre}] - E_{AD} p_{el} + C_{LF}(s - a e_{food}, \delta a) + \pi[c^{cap}(K_R, q_R, Q) + o_R] \quad (21)$$

κ capture of food into the digester, k_{mech} relative methane yield of mechanically separated food, c_{pre} extra pre-treatment (USD/t feed), f_{AD} fugitive methane share, π_{AD} own electricity use, δ digestate landfilled per t feed, π front-end separation charged as a share of the RDF line; e_{food} is the unit vector of the food fraction.

6.5 S4 PHB from landfill gas

$$P_{PHB} = \frac{CH_4^{capt}(s)}{R_{PHB}} \text{ (kg/t)}, \quad T_{PHB} = \frac{P_{PHB} Q_{365}}{10^3}, \quad C_{PHB}(T) = C_{500} \left(\frac{\max(T, 1)}{500} \right)^{-e_{PHB}} \quad (22)$$

$$G_4 = G_{SL} + P_{PHB} (ef_{PHB} - \sigma ef_{PP}), \quad C_4 = C_{SL} + P_{PHB} (C_{PHB} - p_{PHB}) \quad (23)$$

6.6 S5 integrated RDF + AD

$$G_5 = G_{AD}(a) + G_{RDF}(s - a e_{food}) + G_{LF}(rej, \delta a), \quad C_5 = C_{AD}(a) + C_{RDF}(s - a e_{food}) + C_{LF}(rej, \delta a) \quad (24)$$

The whole tonne passes the RDF line, which also separates the food (no separate front-end charge); $rej = (s - a e_{food}) - r$. Mass balance $a + m_{in} + \Sigma rej = 1$ is asserted.

7 Fossil cumulative energy demand and land take

$$PEF = \frac{3.6}{\eta_{pp}} (1 + u_{fuel}) k_{EF} \text{ (MJ/kWh)}, \quad MJ_d = \frac{1 + u_{diesel}}{ef_{diesel}} \text{ (MJ per kg CO}_2\text{e)} \quad (25)$$

$$CED_1 = a_W MJ_d - E_{el} PEF + CED_{LF}, \quad CED_2 = e_R PEF + a_R MJ_d + m_{del} D ef_{tr} MJ_d - 10^3 \psi e_{net} (1 + u_{coal}) + CED_{LF} \quad (26)$$

$$CED_{AD} = a a_A MJ_d - E_{AD} PEF, \quad CED_4 = CED_{SL} + P_{PHB} \left(\frac{ef_{PHB}}{ef_{gas}} - \sigma CED_{PP} \right), \quad CED_{LF} = t a_{LF} MJ_d \quad (27)$$

$$LU_{SL} = \left(\frac{\sum_j m_j}{\rho_{SL} H_{SL}} + \frac{m_{in}}{\rho_{in} H_{SL}} \right) f_{gross}, \quad LU_{OD} = \frac{\sum_j m_j + m_{in}}{\rho_{OD} H_{OD}}, \quad LU_{plant} = \frac{FP}{A_{365} n} \quad (28)$$

Biogenic energy is not counted. Plant land LU_{plant} spreads the footprint FP over the lifetime throughput; each option's land take is the sum of its plants and landfills.

8 Feasibility gates

G1: $H \geq 7$ MJ/kg and G2: $Q \geq 150$ t/day for WtE; G3: $Q \geq 1000$ t/day for the Perpres 109/2025 tariff; G4: $NCV_{RDF} \geq 12.56$ MJ/kg and G5: $D \leq 300$ km for RDF and RDF + AD; G6: $T_{PHB} \geq 500$ t/yr. An option failing a gate is infeasible in that draw.

9 Decision analysis (MSWModel.delta, decide, run_mc)

$$\Delta G_k = G_0 - G_k, \quad \Delta C_k = C_k - C_0, \quad MAC_k = 10^3 \frac{C_k - C_0}{G_0 - G_k} \quad (0 \in \{OD, SL\}) \quad (29)$$

$$CIC_k(p) = C_k + \frac{p G_k}{10^3}, \quad k_i^*(p) = \arg \min_{k \in F_i} CIC_{ik}(p) \quad (30)$$

$$k \text{ Pareto efficient} \Leftrightarrow \text{no } k' : G_{k'} \leq G_k, C_{k'} \leq C_k, (G_{k'}, C_{k'}) \neq (G_k, C_k) \quad (31)$$

$$\hat{P}_k(p) = \frac{1}{N} \sum_{i=1}^N \mathbf{1}[k_i^*(p) = k], \quad SE = \sqrt{\frac{\hat{P}_k(1 - \hat{P}_k)}{N}} \quad (32)$$

$\mathcal{I}_i$ is the set of options feasible in draw i . CIC is the carbon-inclusive cost: it values greenhouse gases only. The carbon value is fixed at $p = 0, 2, 25, 50$ and 100 USD/t CO_2e (policy choice; $2 = \text{UU 7/2021}$ carbon tax). For two options with straight CIC lines the crossing carbon value is $p^* = 10^3(C_b - C_a)/(G_a - G_b)$.

10 Uncertainty propagation

$$x = F_{\text{tri}}^{-1}(u; a, c, b) = \begin{cases} a + \sqrt{u(b-a)(c-a)} & \text{if } u < \frac{c-a}{b-a}; \\ b - \sqrt{(1-u)(b-a)(b-c)} & \text{otherwise} \end{cases}, \quad u \sim U(0, 1) \quad (33)$$

$$\mathbf{S} \sim \text{Dir}(\alpha_0 \mathbf{s} + \boldsymbol{\varepsilon}), \quad S_j = \frac{\Gamma_j}{\sum_j \Gamma_j}, \quad \Gamma_j \sim \text{Gamma}(\alpha_0 S_j + \varepsilon, 1) \quad (34)$$

Every register, CED/land and realism parameter is triangular (low, central, high). Composition is Dirichlet around the harmonised shares with heuristic concentration α_0 (80, or 40 for flagged data); $\varepsilon = 0$ except in the pseudocount scenario (0.5). Each location uses its own random stream, seeded identically in every scenario (common random numbers), so differences between scenarios are paired. Scenarios replace parameter values (e.g. $k_{\text{mech}} = 1$, $c_{\text{pre}} = 0$) or the moisture vector (IPCC values; upper bounds) without changing the equations.

11 Sensitivity analysis

$$\rho_S(X, Y) = \text{corr}(\text{rank}(X), \text{rank}(Y)) \quad (\text{mean } |\rho_S| \text{ over locations}) \quad (35)$$

$$S_{1,x} = \frac{\text{Var}_x[\text{E}(Y|x)]}{\text{Var}(Y)}, \quad S_{T,x} = \frac{\text{E}_x[\text{Var}(Y|x)]}{\text{Var}(Y)} \quad (36)$$

Sobol indices (discovery.sobol_indices) are estimated with Saltelli sampling and the SALib estimators (Saltelli et al. 2010 for S_1 , Jansen for S_T), $N = 512$ base samples, composition fixed at each location. Outputs include the carbon-inclusive cost gap $Y = \text{CIC}_{\text{SS}}(50) - \text{CIC}_{\text{SL}}(50)$.

12 Scenario discovery (mswpath/discovery.py)

$$\hat{\alpha}_0 = \text{median}_j \left[\frac{\bar{s}_j(1 - \bar{s}_j)}{\text{Var}(\bar{s}_j)} - 1 \right] \quad (\text{method of moments over the 21 compositions}) \quad (37)$$

$$\text{Gini}(\text{node}) = 1 - \sum_k \pi_k^2, \quad \text{split} = \arg \max \left[\text{Gini}(\text{parent}) - \sum_c \frac{n_c}{n} \text{Gini}(c) \right] \quad (38)$$

$$\text{coverage} = \frac{\#\{i \in B : k_i^* = k\}}{\#\{i : k_i^* = k\}}, \quad \text{density} = \frac{\#\{i \in B : k_i^* = k\}}{\#\{i \in B\}}, \quad \text{mass} = \frac{\#\{i \in B\}}{N} \quad (39)$$

40,000 condition sets are sampled (carbon value, tonnage, kiln distance, grid, tariff, source separation, composition, moisture, gas collection, landfill cost); the best option of each is explained by a CART tree (depth 4, Gini impurity) and PRIM boxes B , peeled by removing an $\alpha = 5\%$ quantile of one condition (or fixing a binary condition) at a time to maximise density while coverage stays above a threshold.

13 Value of information (mswpath/voi.py)

$$\text{NB}_{ik} = -\text{CIC}_{ik}(p), \quad \text{EVPI} = \frac{1}{N} \sum_i \max_k \text{NB}_{ik} - \max_k \frac{1}{N} \sum_i \text{NB}_{ik} \quad (40)$$

$$\text{EVPPI}(X) = \frac{1}{N} \sum_i \max_k \hat{g}_k(X_i) - \max_k \frac{1}{N} \sum_i \text{NB}_{ik}, \quad \hat{g}_k(X) \approx \text{E}[\text{NB}_k | X] \quad (41)$$

$\hat{g}_k$ is a least-squares regression of NB_k on X (cubic polynomial for one input; additive quadratic for a group of inputs resolved by one measurement campaign), after Strong et al. (2014). The EVPPI of a random input unrelated to the model is subtracted as a bias floor. Options feasible in fewer than half of the draws are removed. Value per year = $\text{EVPPI} \times 365 \text{ Q}_{2025}$.

14 Break-even targets (mswpath/thresholds.py)

$$\Delta(x) = \text{CIC}_k(p; x) - \text{CIC}_{\text{SL}}(p; x), \quad x^* : \Delta(x^*) = 0, \quad x^* = x_a - \Delta(x_a) \frac{x_b - x_a}{\Delta(x_b) - \Delta(x_a)} \quad (42)$$

One input x is swept over a grid with all others central; x^* is found by linear interpolation between the grid points where Δ changes sign; 'never' / 'always' when Δ keeps one sign over the whole range.

15 Projection to 2045 (mswpath/projection.py)

$$EF_{grid}(t) = EF_{grid}(2025) \max[0, 1 - \delta(t - 2025)], \quad \delta = \frac{1}{35} \text{ yr}^{-1} \Rightarrow EF_{grid}(2045) = 0.429 EF_{grid}(2025) \quad (43)$$

$$x(t) = x(2025) (1 + e_x)^{t-2025} : \quad p_{el} (e = 0.02), \quad o_W, o_R, o_A (e = 0.01), \quad c_{SL}, c_{OD} (e = 0.02) \quad (44)$$

$$Q(t) = Q(2025) (1 + g)^{t-2025}, \quad \Delta Y_k = Y_k(2045) - Y_k(2025), \quad \Delta Y_k = \sum_d \Delta Y_k^{(d)} + I_k \quad (45)$$

All equations of Sections 3–9 are re-evaluated with these inputs; CAPEX is constant in real terms and composition, landfill gas and kiln coal are unchanged. $\Delta Y^{(d)}$ is the change with driver d (grid, costs, tonnage) applied alone and I_k the interaction. Local parameter values of the one-city tool are read as 2025 values and escalated in the same way.

16 Symbols and values

16.1 Constants (data/model_constants.csv)

Key	Value	Unit	Meaning	Source
GWP100_CH4	27	-	GWP100 of non-fossil CH4	IPCC AR6 WG1 Table 7.15
GWP100_N2O	273	-	GWP100 of N2O	IPCC AR6 WG1 Table 7.15
GWP20_CH4	79.7	-	GWP20 of non-fossil CH4	IPCC AR6 WG1 Table 7.15
GWP20_N2O	273	-	N2O kept at GWP100 value in the GWP20 case	IPCC AR6 WG1 Table 7.15
LAM	2.443	MJ/kg water	Latent heat of vaporisation	Physical constant
RHO_CH4	0.716	kg/Nm3	Density of methane	Physical constant
LHV_CH4	35.9	MJ/Nm3	Lower heating value of methane	Physical constant
AVAIL	0.85	-	Plant availability (nameplate = throughput / 0.85)	Analyst assumption
P_EL_PSEL	0.2	USD/kWh	Perpres 109/2025 WtE tariff	Perpres 109/2025
PC_MAX	100	USD/t CO2e	Highest carbon value sampled	High-Level Commission on Carbon Prices 2017
QREF_wte	1000	t/d	Reference capacity WtE	Azis et al. 2021
QREF_rdf	300	t/d	Reference capacity RDF	GIZ ERIC-DKTI 2023
QREF_ad	100	t/d	Reference capacity AD	GIZ 2017 in UNDP 2021
QREF_sl	500	t/d	Reference capacity sanitary landfill	World Bank 2024
GATE_lhv_min	7	MJ/kg	G1 WtE minimum LHV as received	Rand et al. 2000
GATE_q_wte	150	t/d	G2 WtE minimum supply	Rand et al. 2000
GATE_q_psel	1000	t/d	G3 PSEL tariff eligibility	Perpres 109/2025
GATE_ncv_rdf	12.56	MJ/kg	G4 RDF minimum NCV (3,000 kcal/kg)	GIZ ERIC-DKTI 2023
GATE_d_max	300	km	G5 maximum road distance to kiln	Analyst assumption
GATE_phb_min	500	t PHB/yr	G6 PHB minimum costed scale	Chidambarampadmavathy et al. 2017
SEED	42	-	Random seed	Model setting
N_MC	4000	-	Monte Carlo draws per location and scenario	Model setting
BASE_YEAR	2025	-	Baseline year	Study design

16.2 Grid emission factors (data/grid_emission_factors.csv)

Grid	EF 2025 (kg CO2/kWh)	EF 2045 (projection)
Jamali	0.877	0.376
Sumatera	0.832	0.357
Mahakam	1.128	0.483

16.3 Fraction properties (data/table2_model_fraction_parameters.csv)

Fraction	w as received (range)	w IPCC	DOC dry	DOCf	C dry	fossil ϕ	LHV dry MJ/kg	τ to RDF
food	0.75 (0.65-0.82)	0.60	0.38	0.70	0.38	0.00	16.5	0.10
garden	0.55 (0.40-0.65)	0.60	0.49	0.70	0.49	0.00	16.0	0.40
paper	0.35 (0.15-0.50)	0.10	0.44	0.50	0.46	0.01	16.0	0.70
wood	0.25 (0.15-0.40)	0.15	0.50	0.10	0.50	0.00	18.0	0.60
textile	0.35 (0.15-0.50)	0.20	0.30	0.50	0.50	0.20	18.5	0.70
rubber	0.15 (0.05-0.25)	0.16	0.47	0.00	0.67	0.20	23.0	0.60
plastic	0.22 (0.05-0.35)	0.00	0.00	0.00	0.75	1.00	30.5	0.80
metal	0.05 (0.02-0.08)	0.00	0.00	0.00	0.00	0.00	0.0	0.00
glass	0.02 (0.01-0.03)	0.00	0.00	0.00	0.00	0.00	0.0	0.00
other	0.25 (0.10-0.40)	0.10	0.00	0.00	0.03	1.00	0.0	0.10

16.4 Register parameters (data/assumption_register.csv)

Key	Central (low–high)	Unit	Meaning	St.
g_default	0.0173 (0.0065–0.0254)	1/yr	Tonnage growth rate where the RIPS gives none (median of six RIPS rates)	A
gamma	0.27 (0.04–0.62)	-	Garden share of lumped 'organik' (median and range of 8 locations reporting both)	A
yard	1 (0.5–1)	-	Share of 'kayu' treated as garden where kayu $\geq$ 10% and no leaf category exists	A
alpha_hi	80	-	Dirichlet concentration, unflagged composition (heuristic, not statistically calibrated)	A
alpha_lo	40	-	Dirichlet concentration, flagged composition (heuristic, not statistically calibrated)	A
F	0.5 (0.475–0.525)	-	Methane fraction of landfill gas	L
doc_k	1 (0.8–1.2)	x	Multiplier on DOC x DOCf (default uncertainty +-20%)	L
h_k	0.9 (0.78–1.02)	x	Calibration factor on legacy dry-matter LHV (not applied to the Gotze plastic value)	L

Key	Central (low–high)	Unit	Meaning	St.
h_plastic_k	1 (0.9–1.1)	x	Uncertainty multiplier on plastic dry LHV (aggregate median, resin mix unknown)	A
efg_k	1 (0.7–1.1)	x	Multiplier on grid emission factor (lower bound = decarbonising grid)	A
docf_rubber	0 (0–0.5)	-	DOCf of rubber/leather: gap scenario (0 = not degradable; 0.5 = IPCC 2006 generic default)	A
mcf_od	0.8 (0.4–0.8)	-	Methane correction factor, open dump	L
mcf_sl	1	-	Methane correction factor, managed anaerobic landfill	L
ox	0.1 (0–0.1)	-	Oxidation in cover soil, sanitary landfill	L
cap	0.5 (0.2–0.8)	-	Lifetime landfill-gas collection efficiency, sanitary landfill	A
anc_od	1 (0–3)	kg CO2e/t	Diesel for dump operation	L
anc_sl	5 (2–10)	kg CO2e/t	Diesel and materials for sanitary landfill operation	L
eta_wte	0.18 (0.14–0.22)	-	Net electrical efficiency of grate incinerator on low-LHV waste	L
n2o_wte	0.05 (0.02–0.08)	kg N2O/t	N2O from continuous stoker incineration	L
anc_wte	10 (5–30)	kg CO2e/t	Auxiliary fuel and flue-gas reagents	L
ash	0.25 (0.15–0.3)	t/t	Bottom ash and APC residue sent to landfill	L
tau_k	1 (0.85–1.1)	x	Multiplier on RDF transfer coefficients	L
omega	0.2 (0.15–0.25)	-	Target moisture of RDF (off-taker limit 20-25%)	L
q_dry	3.2 (2.6–4)	GJ/t water	Heat to evaporate water in the dryer (self-supplied by burning RDF)	L
e_rdf	40 (20–70)	kWh/t	Electricity for shredding, screening, baling	A
psi	0.9 (0.8–1)	-	GJ coal displaced per GJ RDF in the kiln	L
ef_coal	96.1 (94.6–101)	kg CO2/GJ	Combustion emission factor of displaced coal (sub-bituminous)	L
ef_truck	0.1 (0.06–0.15)	kg CO2e/t-km	Truck transport of RDF incl. empty return	L
tort	1.3 (1.2–1.5)	x	Road distance / straight-line distance to nearest kiln	A
anc_rdf	3 (1–6)	kg CO2e/t	Diesel for loaders at the RDF plant	A
kappa	0.7 (0.5–0.9)	-	Share of food waste captured into the digester	L
vs_ts	0.87 (0.8–0.92)	-	Volatile solids / total solids of food waste	L
y_ch4	0.36 (0.28–0.44)	Nm3 CH4/kg VS	Methane yield of source-separated food waste realised in a digester (BMP-type parameter, not landfill DOCf)	V
eta_chp	0.36 (0.32–0.4)	-	Electrical efficiency of biogas engine	L
par_ad	0.2 (0.1–0.3)	-	Own electricity use (pulping, mixing, dewatering)	L
fug_ad	0.05 (0.01–0.1)	-	Fugitive methane from the biogas plant (share of methane produced)	L
anc_ad	3 (1–6)	kg CO2e/t	Diesel for loaders at the AD plant (per t feed)	A
pre	0.4 (0.25–0.6)	-	Front-end separation of food from mixed waste: share of RDF-line cost and electricity	A
dig	0.2 (0.1–0.35)	t/t feed	Dewatered digestate sent to the residue landfill	A
r_phb	2.3 (2–2.8)	t CH4/t PHB	Methane per tonne PHB (1.79 as carbon + 0.52 for process energy)	L
ef_phb	0.5 (0–1.5)	kg CO2e/kg	Nutrients and extraction chemicals	L
ef_pp	1.9 (1.6–3.4)	kg CO2e/kg	Cradle-to-gate footprint of displaced polypropylene	L
sub_pp	1 (0.7–1)	kg/kg	kg PP displaced per kg PHB	L
r	0.1 (0.08–0.12)	1/yr	Real discount rate, same for all pathways	A
n	20 (15–30)	yr	Economic lifetime, same for all pathways	A
b	0.7 (0.6–0.85)	-	Capacity exponent for CAPEX scaling, same for all units	L
K_wte	120 M (100 M–200 M)	USD @1000 t/d	WtE CAPEX at reference capacity	V
o_wte	27 (21–37)	USD/t	WtE O&M;	V
K_rdf	12.3 M (9 M–16 M)	USD @300 t/d	RDF plant CAPEX at reference capacity	L
o_rdf	18.4 (12–24.5)	USD/t	RDF plant O&M;	L
K_ad	7 M (4 M–14 M)	USD @100 t/d	AD plant CAPEX at reference feed capacity	L
o_ad	14 (10.6–30)	USD/t feed	AD plant O&M;	L
c_sl	20 (9–30)	USD/t @500 t/d	Full cost of sanitary landfill with gas collection and flare	L
e_sl	0.3 (0.2–0.4)	-	Scale elasticity of landfill cost per tonne	L
c_od	4 (2–8)	USD/t	Cost of operating an open dump	A
c_phb	6.8 (4.8–9.6)	USD/kg @500 t/a	PHB production cost excl. methane feedstock	A
e_phb	0.083 (0.05–0.15)	-	Scale elasticity of PHB unit cost	A
p_el	0.07 (0.05–0.1)	USD/kWh	Market value of electricity (grid generation cost)	V
p_rdf	1.15 (0.58–1.34)	USD/GJ	Price of RDF at kiln gate	L
p_phb	4 (2.5–6)	USD/kg	Selling price of PHB	L
c_truck	0.1 (0.05–0.15)	USD/t-km	Cost of trucking RDF	A

16.5 Realism, scenario and stress parameters (data/scenario_parameters.csv)

Key	Central (low–high)	Unit	Meaning	Use
y_pen_mech	0.6 (0.35–0.85)	-	Methane yield of food separated mechanically from mixed waste, relative to source-separated food	main case
pre_ofmsw	5 (0–10)	USD/t feed	Extra pre-treatment of mechanically separated food (depackaging, removal of plastics and grit)	main case
glass_imp	0	-	Glass share imputed where glass is not reported (main case: none, i.e. treated as 0)	main case
glass_imp_dist	0.012 (0.0069–0.0448)	-	Distribution used by the scenario glass_imputed (median, p10, p90 of the 15 locations that report glass separately)	scenario
pseudo	0	-	Dirichlet pseudocount added to every fraction (main case 0)	main case
pseudo_scen	0.5	-	Pseudocount used by the scenario dirichlet_pseudocount	scenario
rdf_ncv_k	1	x	Multiplier on the energy content of the delivered RDF (main case 1)	main case
rdf_ncv_k_scen	0.78	x	Stress test: RDF energy scaled so that its NCV falls to about 13-14 MJ/kg (lower than the 15-16.7 MJ/kg reported for Cilacap)	scenario

16.6 CED and land-use factors (data/lcia_factors.csv)

Key	Central (low–high)	Unit	Meaning	St.
pef_eta	0.32 (0.28–0.36)	-	Net thermal efficiency of the fossil generation displaced or consumed (primary energy = 3.6 / eta MJ per kWh)	L
up_fuel	0.08 (0.03–0.15)	-	Upstream supply energy of power-plant fuel (mining, processing, transport) as a share of fuel energy	A
ef_diesel	0.0741	kg CO2/MJ	Converts diesel-type ancillary burdens (kg CO2e) to diesel energy; the ancillary terms also contain materials, so this is an approximation	L
up_diesel	0.15 (0.1–0.25)	-	Upstream (refinery, crude supply) energy of diesel as a share of its energy	A
up_coal	0.05 (0.02–0.1)	-	Upstream (mining, transport) energy of the kiln coal displaced by RDF	A
ef_natgas	0.0561	kg CO2/MJ	Converts the PHB nutrient/chemical burden (kg CO2e) to fossil MJ as if natural-gas based (proxy)	A
ced_pp	75 (65–85)	MJ/kg	Cradle-to-gate fossil CED of displaced polypropylene incl. feedstock energy	L
rho_sl	0.8 (0.6–1)	t/m3	In-place density of compacted MSW in a sanitary landfill	A
h_sl	20 (10–30)	m	Effective fill height of a sanitary landfill	A
f_gross_sl	1.3 (1.1–1.5)	-	Gross site area / fill area (roads, leachate ponds, buffer)	A
rho_inert	1.3 (1–1.6)	t/m3	In-place density of ash, digestate cake and other inert residues	A
rho_od	0.5 (0.35–0.7)	t/m3	Density of uncompacted waste in an open dump	A
h_od	5 (3–10)	m	Average heap height of an open dump	A
fp_wte	40 (20–80)	m2 per t/d	Site footprint of a WtE plant per t/d nameplate capacity	A
fp_rdf	50 (25–100)	m2 per t/d	Site footprint of an RDF plant (incl. drying and storage) per t/d	A
fp_ad	60 (30–120)	m2 per t/d	Site footprint of an AD plant (incl. digestate handling) per t/d feed	A

16.7 Projection parameters (data/projection_2045_parameters.csv)

Key	Value	Unit	Meaning
base_year	2025	year	Year of the existing (baseline) condition
target_year	2045	year	Projection year
netzero_year	2060	year	Year in which the grid emission factor reaches zero
delta_grid	0.028571428571428571	1/yr	Linear decline of the grid emission factor per year, delta = 1/35 (2025 to 2060): EF(t) = EF2025 x max(0, 1 - delta (t - 2025)); applied to GHG and fossil CED of grid electricity
esc_electricity	0.02	1/yr	Real escalation of the market value of electricity (revenue of WtE and AD)
esc_labour	0.01	1/yr	Real escalation of plant operation and maintenance cost (labour-driven O&M;)
esc_landfill	0.02	1/yr	Real escalation of landfill and open-dump cost (third rate of the 2/1/2% set; interpretation to be confirmed)
capex_escalation	0	1/yr	CAPEX held constant in real terms
tonnage_growth	location	1/yr	Tonnage grows from 2025 to 2045 at each location's growth rate (RIPS rate or 1.73%/yr default); composition held constant (no defensible composition trend)

16.8 Symbol map: equations to data keys

Symbol	Data key
$k_D, F, MCF_{SL}, MCF_{OD}, \eta, OX$	doc_k, F, mcf_sl, mcf_od, cap, ox
$a_{LF}, c_{SL}, e_{SL}, c_{OD}$	anc_sl / anc_od, c_sl, e_sl, c_od
$k_h, k_{h,p}, DOCf_R, k_{EF}$	h_k, h_plastic_k, docf_rubber, efg_k
$\eta_W, \eta_{N2O}, a_W, \alpha, K_W, o_W$	eta_wte, n2o_wte, anc_wte, ash, K_wte, o_wte
$k_t, \omega, q_{dry}, e_R, \psi, EF_{coal}, ef_{tr}, a_R, K_R, o_R, p_{RDF}, c_{tr}$	tau_k, omega, q_dry, e_rdf, psi, ef_coal, ef_truck, anc_rdf, K_rdf, o_rdf, p_rdf, c_truck
$\kappa, VS/TS, y_{CH4}, k_{mech}, c_{pre}, f_{AD}, \eta_{CHP}, \pi_{AD}, \delta, \pi, K_A, o_A$	kappa, vs_ts, y_ch4, y_pen_mech, pre_ofmsw, fug_ad, eta_chp, par_ad, dig, pre, K_ad, o_ad
$R_{PHB}, c_{500}, e_{PHB}, p_{PHB}, ef_{PHB}, ef_{PP}, \sigma$	r_phb, c_phb, e_phb, p_phb, ef_phb, ef_pp, sub_pp
r, n, b, A, q_{ref}	r, n, b, AVAIL, QREF_*

Symbol	Data key
η_{pp} , u_{fuel} , u_{diesel} , ef_{diesel} , u_{coal} , CED_{pp} , ρ , H , f_{gross} , FP	<code>pef_eta</code> , <code>up_fuel</code> , <code>up_diesel</code> , <code>ef_diesel</code> , <code>up_coal</code> , <code>ced_pp</code> , <code>rho_*</code> , <code>h_*</code> , <code>f_gross_sl</code> , <code>fp_*</code>
γ , γ_0 , α_0	<code>gamma</code> , <code>yard</code> , <code>g_default</code> , <code>alpha_hi / alpha_lo</code>

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- Full source list of every parameter: [data/assumption_register.csv](#), [data/secondary_data_verification.csv](#) and [MSW_Methodology_v3_EN.pdf](#).